
SWS3009 – Part B

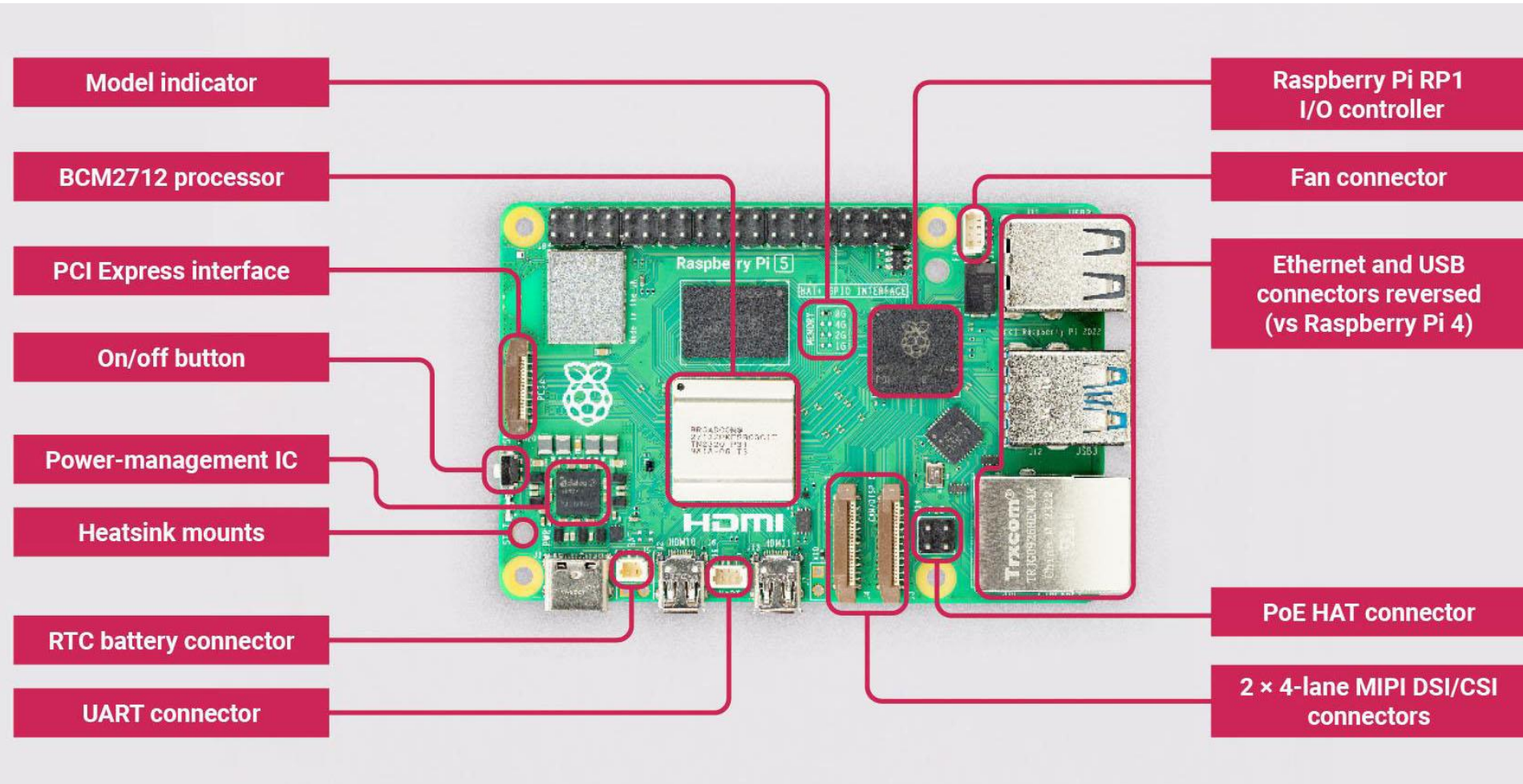
Robotics / TeleOperation

Section 2: Raspberry Pi

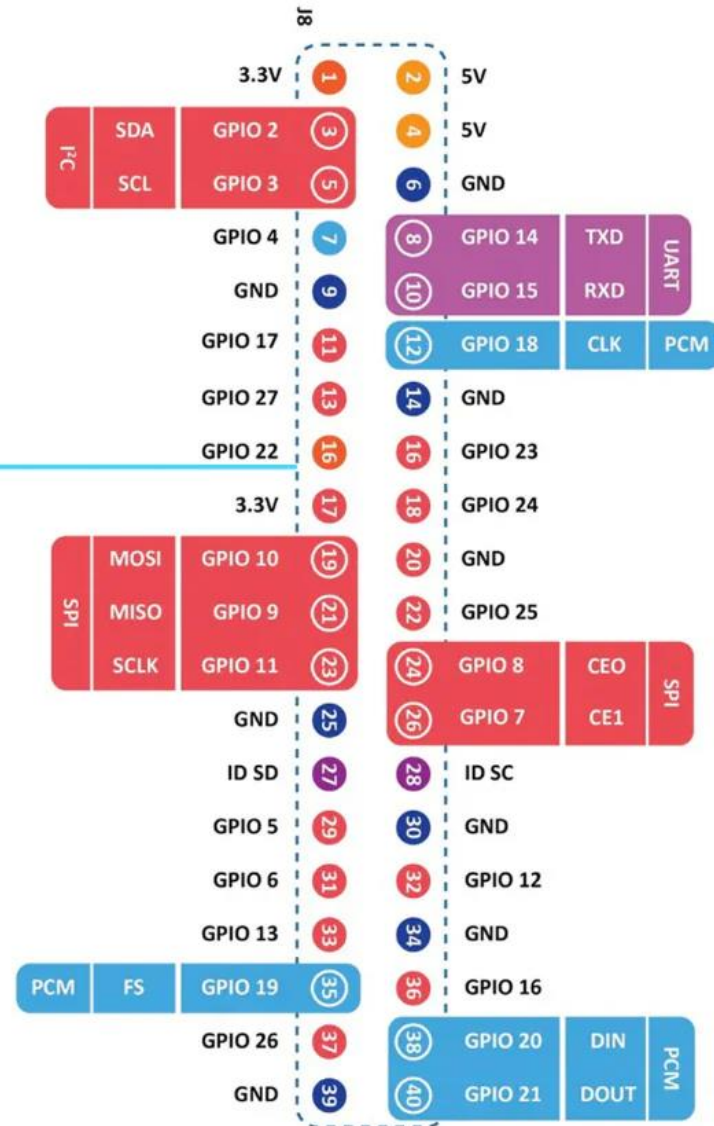
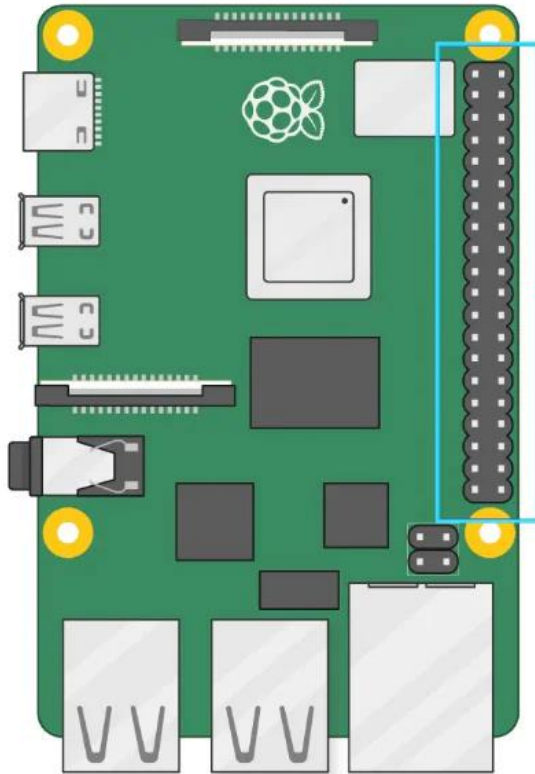
Raspberry Pi, Apple Pi, Banana Pi,

PI BASICS

Raspberry Pi 5 – Layout



Raspberry Pi 5 – Layout



Key Characteristics

- Pi is simply a **tiny computer**
 - ❑ **Processor** = Quad-Core ARM Processor
 - ❑ **Memory** = Built-in
 - ❑ **Hard disk** = SD Card (16Gb in our case)
 - ❑ **Operating System** = Commonly Linux based (**Raspbian** in our case)
 - ❑ **I/O interfaces** = HDMI, USB, Audio etc
- Useful functionalities for us:
 - ❑ **Networking**: Wifi, Lan
 - ❑ **Interfaces**: GPIO, Camera

And then.....?

- Raspberry Pi is essentially a miniscule computer
- Instead of “talking about it”, we are going to just get stuck in and try it out!

Your Tasks for the Morning

Follow the sheet first!

- Get your RPi up and going (headless)
 - ssh and vnc (will require some googling)
- Download a game or piece of software
- Take a photo or a video using the PiCamera
(you can see the python code provided)

Reference

- Pi Camera, "Getting Started"
 - <https://projects.raspberrypi.org/en/projects/getting-started-with-picamera>

END